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ORIGINAL ARTICLE

Dynamics and model representation of the boreal spring subseasonal variability over North America

Jaeyoung Hwang[1] · Zhenyu You[1] · Jie He[1] · Yi Deng[1]

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Abstract

The dynamics and model representation of boreal spring subseasonal variability over North America are examined, focusing on four leading modes identified through empirical orthogonal functions. Mode 1 is active over the high latitudes of the North Pacific, exhibiting a meridional dipole structure. Mode 2 features a localized circulation anomaly over northeastern North America. Modes 3 and 4 are wave-train-like patterns extending from the eastern North Pacific to North America from lower- and mid-latitudes, respectively. Despite a bias in Mode 2, the Community Atmosphere Model version 5 (CAM5) reproduces the overall spatiotemporal features and associated temperature and precipitation responses to the subseasonal modes. A scale-resolving quasi-geostrophic geopotential tendency analysis reveals that Modes 1 and 2 develop quasi-stationarily followed by westward movements, primarily driven by vorticity fluxes. Heat fluxes act to maintain the barotropic structure of the related circulation anomalies. Modes 3 and 4 exhibit eastward propagation throughout their lifecycles, explained by both vorticity and heat fluxes. Scale separation indicates that both high- and low-frequency eddy vorticity fluxes explain the formation of Modes 1 and 2, whereas the latter dominates the propagating nature of Modes 3 and 4. These dynamical characteristics are well replicated in CAM5. The bias in Mode 2 is likely attributed to weaker contributions of high-frequency components in CAM5. Further analyses suggest that current models in the Coupled Model Intercomparison Project Phase 6 also reasonably capture the basic statistics of the subseasonal modes, lending confidence in using them for the study and prediction of long-lasting weather and climate extremes.

Keywords Subseasonal variability · North America · CAM5 · Quasi-geostrophic geopotential tendency analysis

1 Introduction

Recent decades have witnessed an increasing frequency of weather extremes in North America (Brown et al. 2010; Diffenbaugh and Ashfaq 2010; Coumou and Rahmstorf 2012; Stevenson and Schumacher 2014; Swain et al. 2018; Thompson et al. 2022; Picard et al. 2023). Subseasonal (or intraseasonal) variability, typically referring to atmospheric variability with periods ranging from weeks to a season, is considered as one of the important factors modulating the occurrence of North American weather extremes. For instance, the persistent cold advection downstream of a subseasonal-scale anticyclone gives rise to cold extremes in boreal winter (e.g., Harnik et al. 2016; Xie et al. 2017; Park et al. 2024; Hwang et al. 2024). In summer, a clear sky with enhanced downward short-wave radiation over an anticyclonic circulation center increases the daily maximum surface temperature in the region (e.g., Pfhal and Wernli 2012; Dong et al. 2018; White et al. 2023; Coumou et al. 2014). Subseasonal circulation anomalies are also reported to be associated with hydrological extremes such as floods and droughts across the season (e.g., Dong et al. 2011; Payne and Magnusdottir 2014; Dong et al. 2018; Hicks et al. 2023; Dong et al. 2024).

In general, subseasonal modulation of springtime extremes has received relatively less attention due to relatively rare occurrences of temperature extremes in a transition season. However, this modulation could be important in boreal spring over North America, where the subseasonal

✉ Jaeyoung Hwang
jhwang362@gatech.edu

✉ Yi Deng
yi.deng@eas.gatech.edu

[1] School of Earth and Atmospheric Sciences, Georgia Institute of Technology, 311 Ferst Drive, Atlanta, GA 30332-0340, United States of America

variability is coupled with intense synoptic and mesoscale disturbances (e.g., Hoskins and Hodges 2019; Feng et al. 2019; You and Deng 2023). For instance, subseasonal variability could create favorable conditions for heavy precipitation by affecting the low-level moisture transport (e.g., Munoz and Enfield 2011; Lee et al. 2016) and/or enhancing the large-scale upward motion through upper-level divergence (e.g., Song et al. 2019). It may also facilitate the development of synoptic and mesoscale weather systems and change their trajectories (e.g., Grise et al. 2013; Hicks et al. 2023), thus controlling the local hydrology.

There have been many efforts to understand atmospheric subseasonal variability over North America. These efforts fall into two broad categories. One targets localized high-latitude variabilities such as atmospheric blocking (Dole 1986; Woollings et al. 2018; Lupo 2021), North Atlantic Oscillation (Hurrell et al. 2003), and North Pacific Oscillation (Linkin and Nigam 2008). The other focuses on wave patterns originating from the tropics (e.g., teleconnections of the Madden Julian Oscillation and the Pacific North America patterns; Wallace and Gutzler 1981; Stan et al. 2017; Barnston and Livezey 1987) or zonally oriented wave patterns in the mid-latitudes (e.g., Ding and Wang 2005; Coumou et al. 2014; Song et al. 2021). Modes of subseasonal variabilities have also been referred to as weather regimes or persistent anomalies (Reinhold and Pierrehumbert 1982; Legras and Ghil 1985; Tuel and Martius 2023; Lee et al. 2023).

Given its scales, subseasonal variability has been considered as an important source of predictability for medium to long-term range forecasting of extremes in operational and climate models (e.g., Kim et al. 2023; Baggett et al. 2017; Vigaud et al. 2018; Sung et al. 2019). The difficulty in representing small-scale features such as fronts or mesoscale systems in global models further fuels the need for using modes of subseasonal variability as indirect indicators of extremes (e.g., Song et al. 2022). This inquiry has naturally led to many studies on the representation of subseasonal variability in models where robust biases have been found to persist (Chen et al. 2018; Kim et al. 2023; Davini and Andrea 2020). Most studies on subseasonal variability in models have focused on the winter and summer seasons, and less attention was placed on subseasonal variability in transitional seasons such as spring. There is a limited understanding of how well models simulate the characteristics and dynamics of spring subseasonal variability over North America and how the surface impacts of important modes of subseasonal variability are represented in models.

Against such a backdrop, the present study aims to obtain a comprehensive picture of the spring subseasonal variability over North America. Specifically, we examine the leading modes of subseasonal variability in terms of their characteristics, precipitation impacts, and formation dynamics in global models, including the Community Atmosphere Model version 5 (CAM5) and the Coupled Model Intercomparison Project Phase 6 (CMIP6). Following this introduction, the data, model experiments, and quasi-geostrophic geopotential tendency diagnosis are described in Sect. 2. The statistics and dynamics of the subseasonal modes and their model representation are presented in Sect. 3. Finally, a summary and discussion are provided in Sect. 4.

2 Data and methods

2.1 Data

The European Centre for Medium-Range Weather Forecasts Interim reanalysis (ERA-I; Dee et al. 2011) is adopted for the period of 1979–2014. To understand the dynamics of modes of North American subseasonal variability, instantaneous 6-hourly variables across 13 pressure levels are analyzed. Global gauge-based daily precipitation data from the Climate Prediction Center (CPC; Xie et al. 2007) is also utilized to examine the surface impact of subseasonal variability in North America. All variables are interpolated to a 2.5° by 2.5° grid and examined during boreal spring (March–April–May).

To assess the model representation of the modes of North American subseasonal variability in a global model, a global ocean–global atmosphere experiment of CAM5 (CAM5-GOGA) is utilized. The CAM5-GOGA was performed with boundary conditions from the observed monthly time-varying sea surface temperature (ERSST4; Extended Reconstructed Sea Surface Temperature version 4) and sea ice (HadISST1; Hadley Centre Sea Ice and Sea Surface Temperature version 1) at a 0.9° by 1.25° horizontal resolution. To understand the uncertainty derived from internal variability (e.g., Deser et al. 2012; Kay et al. 2015), 10 ensemble members of CAM5-GOGA are analyzed in this study. Since 6-hourly data are unavailable in CAM5-GOGA, we additionally conducted a similar experiment using different monthly time-varying sea surface temperature (SST) and sea ice, both from the merged Hadley-OI dataset (HadISST and NOAA-Optimal Interpolation; Hurrell et al. 2008). Although different SST and sea-ice datasets are prescribed, the simulation exhibits overall similar results compared to CAM5-GOGA (not shown). Historical simulations (1979–2014) from 20 models archived in the CMIP6 (Eyring et al. 2016) are further analyzed to examine their general representations of springtime North American subseasonal variability (Table 1).

Table 1 Model data description for CMIP6

Model	Lat × Lon	Ensemble	Experiment	Institution
ACCESS-CM2	1.3° × 1.9°	r1i1p1f1	Historical	CSIRO
BCC-CSM2-MR	1.1° × 1.1°	r1i1p1f1	Historical	BCC
BCC-ESM1	2.8° × 2.8°	r1i1p1f1	Historical	BCC
CanESM5	2.8° × 2.8°	r1i1p1f1	Historical	CCCma
CESM2	0.9° × 1.3°	r1i1p1f1	Historical	NCAR
CESM2-FV2	1.9° × 2.5°	r1i1p1f1	Historical	NCAR
CESM2-WACCM-FV2	1.9° × 2.5°	r1i1p1f1	Historical	NCAR
FGOALS-f3-L	1.0° × 1.25°	r1i1p1f1	Historical	CAS
FGOALS-g3	2.3° × 2.0°	r1i1p1f1	Historical	CAS
GFDL-CM4	1.0° × 1.25°	r1i1p1f1	Historical	NOAA-GFDL
INM-CM4-8	1.5° × 2.0°	r1i1p1f1	Historical	INM
INM-CM5-0	1.5° × 2.0°	r1i1p1f1	Historical	INM
IPSL-CM6A-LR	1.3° × 2.5°	r1i1p1f1	Historical	IPSL
MIROC6	1.4° × 1.41°	r1i1p1f1	Historical	MIROC
MPI-ESM1-2-HR	0.94° × 0.94°	r1i1p1f1	Historical	MPI
MPI-ESM1-2-LR	1.9° × 1.9°	r1i1p1f1	Historical	MPI
MRI-ESM2-0	1.1° × 1.1°	r1i1p1f1	Historical	MRI
NorESM2-LM	1.9° × 2.5°	r1i1p1f1	Historical	NCC
NorESM2-MM	0.9° × 1.3°	r1i1p1f1	Historical	NCC
TaiESM1	1.3° × 1.9°	r1i1p1f1	Historical	AS-RCEC

2.2 Characterization of the leading modes of subseasonal variability

To understand the atmospheric subseasonal variability over North America, we first performed an empirical orthogonal function (EOF) analysis of the 8–30-day bandpass-filtered (e.g., Jiang et al. 2013) geopotential height at 500 hPa (Z500) over the eastern North Pacific to North America sector (170–70° W, 20–70° N). Here, four leading North American subseasonal modes (~60% of total variability) are identified (Fig. 1), which are statistically distinct based on North's criteria (North et al. 1982). The first and second modes (Modes 1 and 2) are characterized by two localized high-latitude circulation anomalies over the North Pacific and northeastern North America (Figs. 1a,b), whereas Modes 3 and 4 assemble the eastward propagating Rossby wave trains across the North Pacific-North America sector (Figs. 1c,d). To identify these subseasonal modes across multiple datasets including reanalysis and climate models, we define the indices for subseasonal modes as linear combinations of 8–30-day filtered Z500 based on the centers of action in each mode. The centers of action in individual modes are identified based on the composite distributions of the 8–30-day filtered Z500 over a larger domain (160°E-40°W, 20°-80°N) when the principal component value exceeds one standard deviation. Since the values of 8–30-day filtered Z500 exhibit magnitudes up to the order of 10^2 across the North Pacific-North America sector, the index is normalized by dividing by 100. Although not shown here, the overall results are not much changed when small adjustments are made to the centers of action.

$$\begin{aligned} Index_{mode1} =& (Z500_{8-30}(205^\circ\mathrm{E},\ 60^\circ\mathrm{N}) \\ & - Z500_{8-30}(207.5^\circ\mathrm{E},\ 32.5^\circ\mathrm{N}) \\ & - Z500_{8-30}(270^\circ\mathrm{E},\ 50^\circ\mathrm{N}))/100 \end{aligned}$$

$$\begin{aligned} Index_{mode2} =& (Z500_{8-30}(275^\circ\mathrm{E},\ 62.5^\circ\mathrm{N}) \\ & - Z500_{8-30}(302.5^\circ\mathrm{E},\ 37.5^\circ\mathrm{N}))/100 \end{aligned}$$

$$\begin{aligned} Index_{mode3} =& (Z500_{8-30}(245^\circ\mathrm{E}, 50^\circ\mathrm{N}) \\ & - Z500_{8-30}(207.5^\circ\mathrm{E}, 45^\circ\mathrm{N}) \\ & - Z500_{8-30}(285^\circ\mathrm{E}, 45^\circ\mathrm{N}))/100 \end{aligned}$$

$$\begin{aligned} Index_{mode4} =& (Z500_{8-30}(225^\circ\mathrm{E}, 47.5^\circ\mathrm{N}) \\ & - Z500_{8-30}(190^\circ\mathrm{E}, 50^\circ\mathrm{N}) \\ & - Z500_{8-30}(250^\circ\mathrm{E}, 35^\circ\mathrm{N}) \\ & + Z500_{8-30}(290^\circ\mathrm{E}, 43.5^\circ\mathrm{N}))/100 \end{aligned}$$

2.2.1 Scale-resolving quasi-geostrophic geopotential tendency diagnosis

To understand the dynamics of the identified subseasonal modes, the quasi-geostrophic (QG) geopotential tendency equation (Hwang et al. 2020) is utilized under the adiabatic and frictionless assumption:

$$L\left(\frac{\partial \phi'}{\partial t}\right) = -\left(\nabla \cdot \boldsymbol{V}\left(\zeta + f\right)\right)' + f_o \frac{\partial}{\partial p}\left(\nabla \cdot \frac{\boldsymbol{V}\theta_d}{s_o}\right)' \quad (1)$$

$$L \equiv \frac{1}{f_o}\nabla^2 + f_o \frac{\partial}{\partial p}\left(\frac{1}{s_o}\frac{p}{R}\left(\frac{p_o}{p}\right)^{\frac{R}{c_p}}\frac{\partial}{\partial p}\right)$$

where, ∇ is a horizontal differential operator; $\boldsymbol{V}$ is horizontal wind vectors; ζ and f are relative and planetary vorticity,

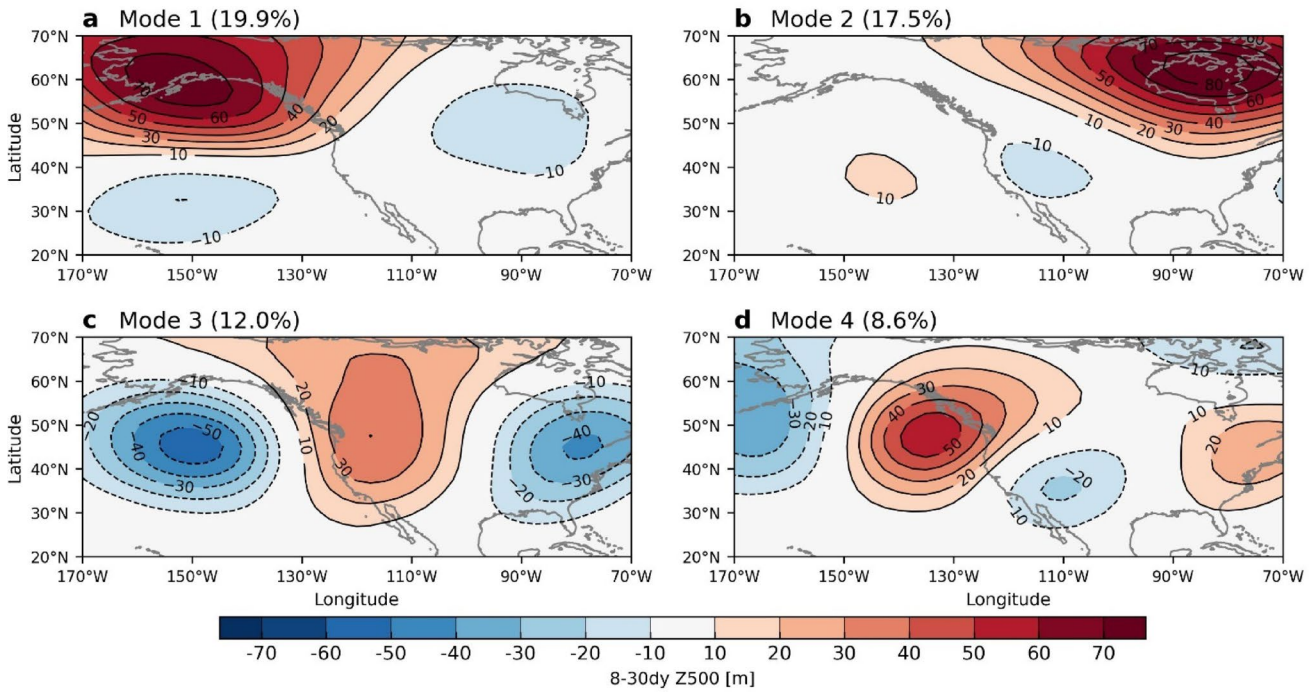


Fig. 1 Four leading modes identified by empirical orthogonal function (EOF) analysis for 8–30-day bandpass-filtered 500-hPa geopotential height (Z500) over the eastern North Pacific-North America region (170–70° W; 20–70° N) during spring. The zero contour is omitted. The explained variance by each EOF mode is indicated in parentheses at the top

respectively; ϕ is geopotential; p is pressure; f_o is set to the Coriolis parameter at 45° N. The gas constant for dry air is R, and the specific heat at constant pressure is denoted as c_p. The reference pressure (p_o) is chosen as 1000 hPa. The hemispheric mean static stability parameter is denoted by $s_o(\equiv -\partial\theta_o/\partial p)$. θ is potential temperature, and θ_d is given as a departure from its hemispheric mean value (θ_o). Prime indicates an anomaly component which is a deviation from the calendar-day climatological value.

The three-dimensional Laplacian (L) is solved from 1000 to 50 hPa using the successive over-relaxation iteration. As a boundary condition, the adiabatic thermodynamic equation is applied at the top and bottom levels (Lau and Holopainen 1984):

$$-\frac{p}{R}\left(\frac{p_o}{p}\right)^{\frac{R}{c_p}}\frac{\partial}{\partial p}\left(\frac{\partial\phi}{\partial t}\right) = -\nabla\cdot\boldsymbol{V}\theta_d \tag{2}$$

To quantify the high-frequency and low-frequency eddy contributions in the dynamics of the subseasonal modes, the vorticity flux divergence in Eq. (1) is decomposed into high- and low-frequency components:

$$-\left(\nabla\cdot\boldsymbol{V}\left(\zeta+f\right)\right)' = HFV + LFV \tag{3}$$

$$HFV = -\left(\nabla\cdot\boldsymbol{V}_H\left(\zeta_H+f\right)+\nabla\cdot\boldsymbol{V}_H\zeta_L+\nabla\cdot\boldsymbol{V}_L\zeta_H\right)'$$
$$LFV = -\left(\nabla\cdot\boldsymbol{V}_L\left(\zeta_L+f\right)\right)'$$

where subscript H indicates an 8-day high-pass filtered component. Subscript L is defined as the difference between total and 8-day high-pass filtered components. Here, low-frequency vorticity forcings (LFV) are defined as the component involving solely low-frequency quantities, whereas high-frequency vorticity forcings (HFV) are defined as the remaining terms in the vorticity flux divergence (e.g., Nakamura et al. 1997; Sung et al. 2021).

3 Results

3.1 Characteristics, surface impacts, and CAM5 representation of the subseasonal modes

The composites of the 8–30-day bandpass-filtered Z500 associated with North American springtime subseasonal modes in there analysis and CAM5-GOGA are displayed in Fig. 2. Here, the composites are constructed using data from days when the daily anomalies of the indices of the subseasonal variability (Methods) exceed one standard deviation in magnitude. Only the positive phase of each subseasonal mode is considered since the results are overall symmetric between the two opposite phases. Mode 1 exhibits a pronounced positive (i.e., anticyclonic) Z500 anomaly in the high-latitude North Pacific (Fig. 2a). On its equatorward side, a relatively weaker cyclonic circulation appears,

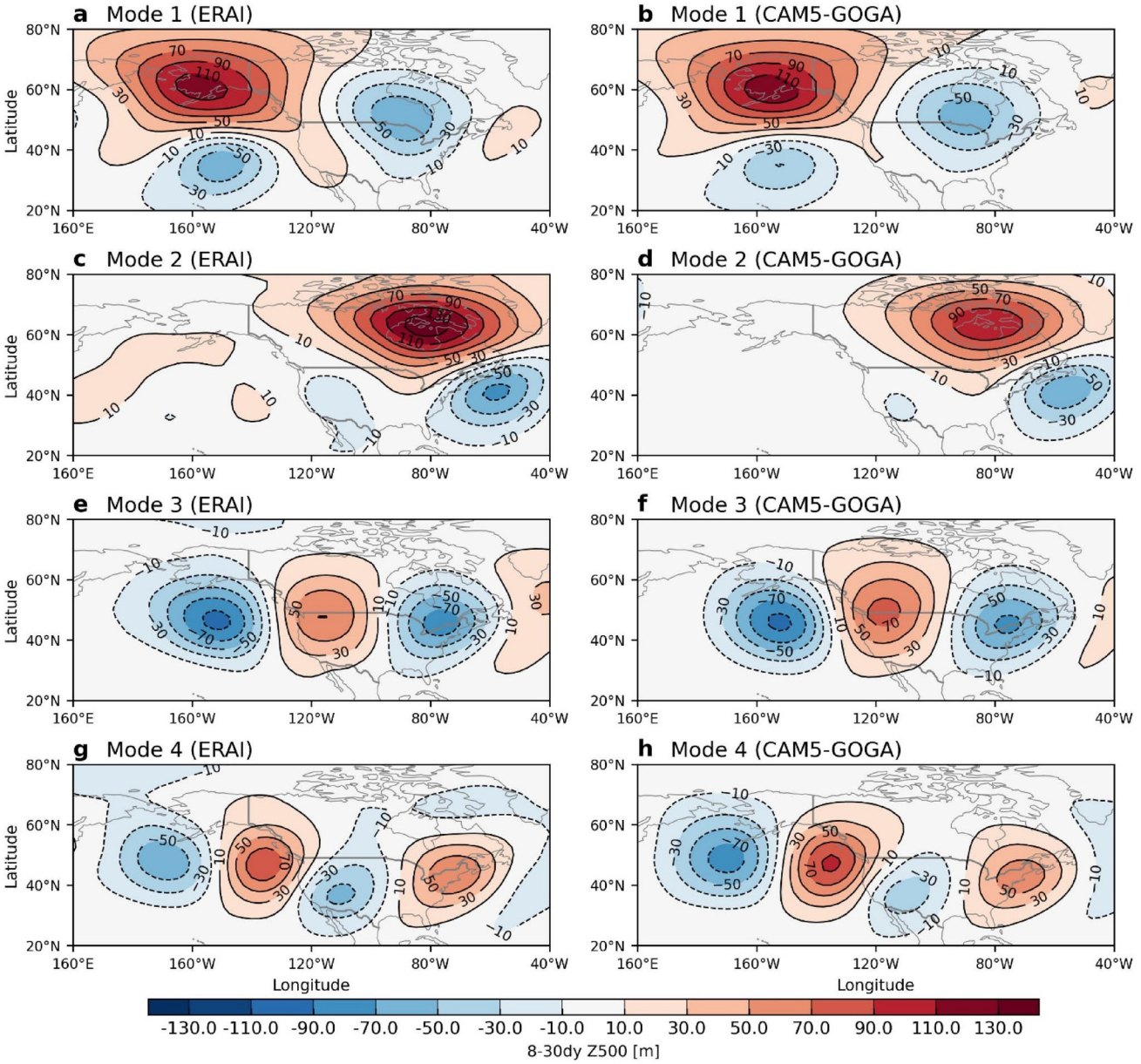


Fig. 2 Composite of the 8–30-day bandpass-filtered Z500 (10 m interval) associated with four subseasonal modes in (left) ERA-I and (right) the ensemble mean of CAM5-GOGA. The composite is performed when the magnitude of the indices, defined by linear combinations of centers of action in each EOF mode (Methods), exceeds its standard deviation. The zero line is omitted

showing a meridional dipole structure. Downstream of the high-latitude circulation, a negative Z500 anomaly also appears, inducing anomalous northerlies over western North America. These spatial patterns also resemble North Pacific blocking patterns (e.g., Park et al. 2024), which give rise to a significant surface impact on North America (e.g., Carrera et al. 2004; Breeden et al. 2020; Hwang et al. 2024). Mode 2 exhibits a localized anticyclonic circulation centered over northeastern North America (Fig. 2c). It is accompanied by cyclonic circulation anomalies over its southwest and southeast flanks, which act to weaken the North Atlantic jet. In contrast, wave-train-like circulation patterns extending from the eastern North Pacific to North America are identified in Modes 3 and 4. Mode 3 is characterized by a tripolar structure of a cyclone-anticyclone-cyclone pair over North America (Fig. 2e), whereas a slightly more compact circulation packet is observed in Mode 4 (Fig. 2g).

The four subseasonal modes in North America are also assessed in CAM5-GOGA. The model successfully reproduces the overall circulation patterns with one notable bias. The model underestimates the high-latitude variability over northeastern North America in Mode 2 (capturing ~ 70% of

the observed maximum amplitude) (Fig. 2d). For Modes 1, 3, and 4, however, minor biases are observed in CAM5-GOGA (Figs. 2b,f,h).

To understand the occurrence characteristics of these subseasonal modes in reanalysis and CAM5-GOGA, probability density functions (PDFs), based on kernel density estimation, of the four subseasonal mode indices are illustrated in Fig. 3. The subseasonal modes in the reanalysis exhibit overall symmetrical distribution (blue lines in Fig. 3). The model remarkably well reproduces the PDFs of the subseasonal mode indices except for Mode 2 (red lines). While Mode 2 in CAM5-GOGA exhibits a symmetrical distribution, it shows a significantly narrower distribution (i.e., higher kurtosis) compared to the reanalysis (Fig. 3b), in agreement with the weaker variability shown in Fig. 2d. The spread of PDFs from individual CAM5-GOGA ensembles (light red lines), which represents uncertainties driven by internal variability (Hawkins and Sutton 2009; Deser et al. 2012; Kay et al. 2015), exhibits a distinct separation from the PDF in the reanalysis for Mode 2. This means that Mode 2 is systematically biased in the model. It contrasts with other modes where the PDFs in the reanalysis generally lie within the ensemble spreads.

How well does the model simulate the surface impact of subseasonal modes? To answer this question, the responses of surface air temperature to the subseasonal modes in the reanalysis and CAM5-GOGA are compared in Fig. 4. In Mode 1, cold anomalies appear around the downstream cyclonic circulation (Fig. 4a). A minimum temperature anomaly is observed over western Canada resulting from cold advection by southward flows. The anticyclonic circulation of Mode 2 induces a warm anomaly over northeastern North America (Fig. 4c), partly due to an increased downward short-wave radiation over an anticyclonic circulation center (e.g., Pfahl and Wernli 2012). Although weak, a cold anomaly is also present over the cyclonic circulation in western United States. In Mode 3, the dipolar surface air temperature anomalies correspond to anticyclonic and cyclonic circulations (Fig. 4e), with a warm anomaly over western North America but a cold anomaly over the east. Similarly, tripolar surface air temperature responses are induced by Mode 4 across North America (Fig. 4g). Warm anomalies appear over the west coast of Canada and eastern North America along the anticyclonic circulations. In contrast, a cold anomaly is driven by the cyclonic circulation centered over the western United States.

The CAM5-GOGA captures well the spatial distribution of temperature responses to subseasonal modes. The pronounced cold anomaly over North America is driven by Mode 1 in the model (Fig. 4b), although its magnitude is slightly weaker than in the reanalysis. The temperature response to Mode 2 is characterized by a warm anomaly over northeastern North America and a cold anomaly over the western United States in the model (Fig. 4d). As in Mode 1, both responses are weaker in the model compared to the reanalysis. In contrast to the reanalysis, the model simulates

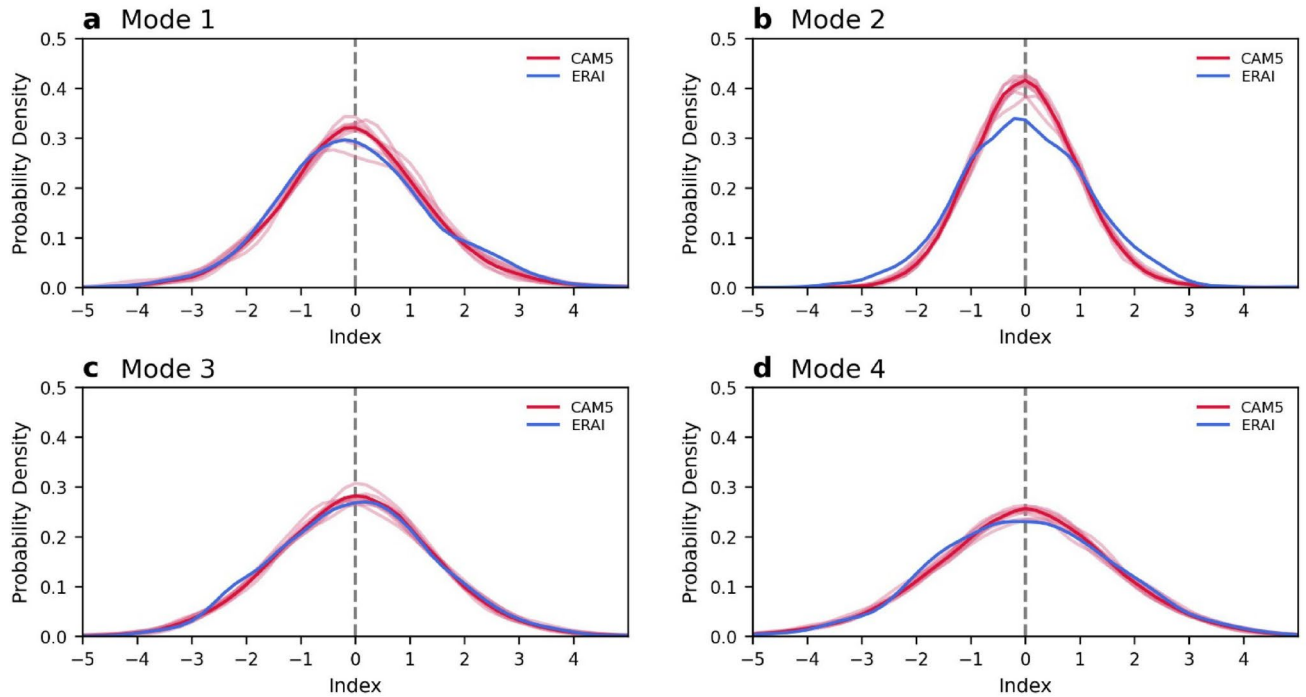


Fig. 3 Probability density function (PDF) of four subseasonal mode indices. Blue and red lines indicate the PDF in ERA-I and the ensemble mean of CAM5-GOGA, respectively. Light red lines illustrate PDFs from individual ensemble members in the CAM5-GOGA simulation

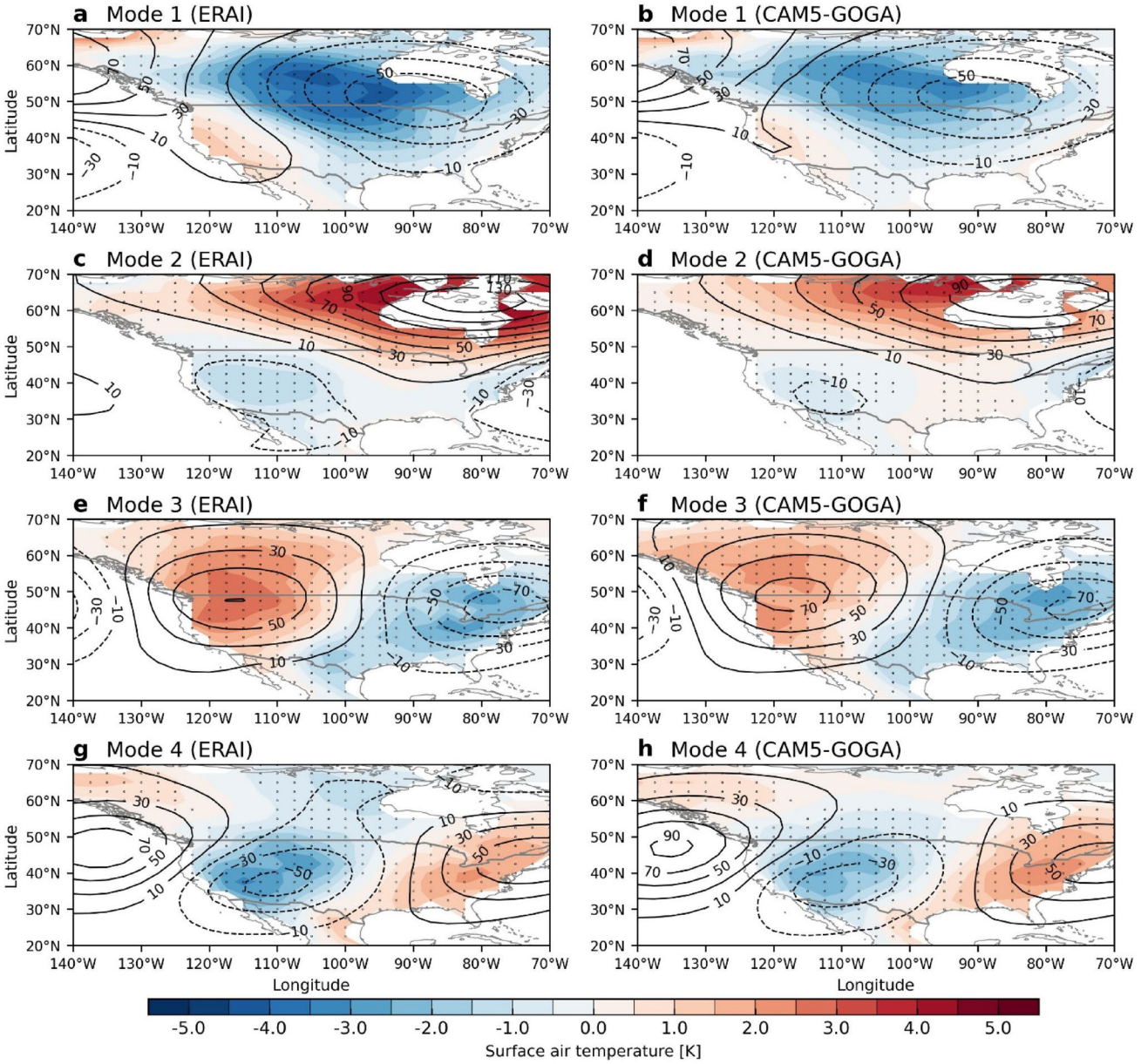


Fig. 4 Composite of the surface air temperature anomaly (K; shading) and Z500 anomaly (10 m interval; contour) associated with four subseasonal modes in (left) ERA-I and (right) the ensemble mean of CAM5-GOGA. The regions with statistical significance at the 95% confidence level are marked with dots. The composite is performed when the magnitude of indices, defined by linear combinations of centers of action in each EOF mode, exceeds its standard deviation

a warm anomaly over the central United States. For Mode 3, the model successfully reproduces the dipolar temperature anomalies across North America, although the response is slightly weaker over the region of the anticyclonic circulation (Fig. 4f). The tripolar temperature responses to Mode 4 are also captured by the model, with a weaker response over the western United States but a stronger response over eastern North America (Fig. 4h).

The same analysis is conducted for daily precipitation anomalies in Fig. 5. In Mode 1, a marked decrease in precipitation is observed along the west coast of North America, due to weakened moisture transport and subsidence indicated by the horizontal convergence (Fig. 5a). The anticyclonic circulation in Mode 2 is associated with a dipole precipitation pattern over the United States, characterized by increased precipitation in upstream regions and decreased precipitation downstream (Fig. 5c). In Mode 3, reduced precipitation is observed across the northern United States and southern Canada (Fig. 5e). However, enhanced precipitation is present along the northwest coast of North

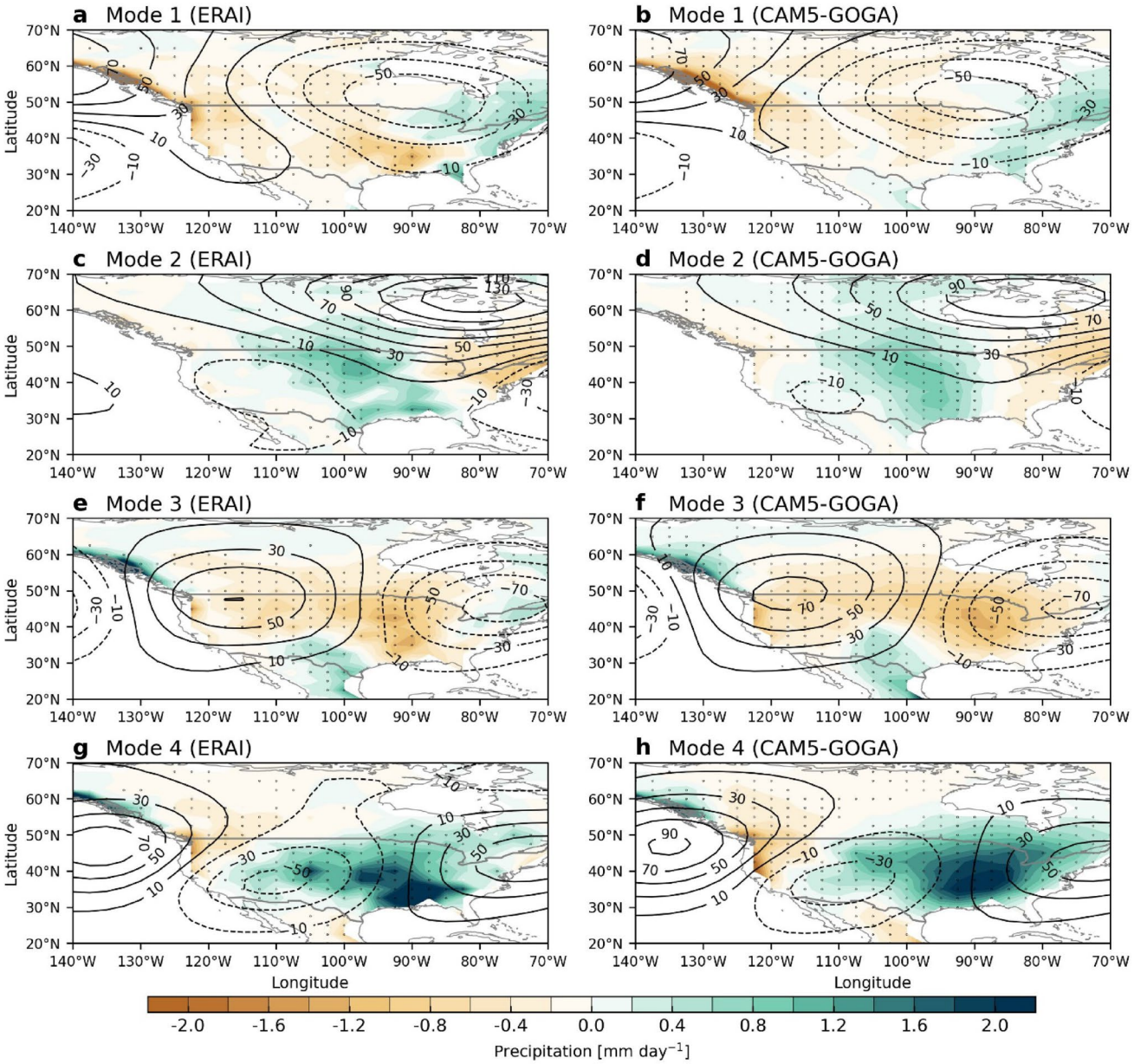


Fig. 5 Same as Fig. 4 but for precipitation anomaly (mm day^{-1})

America. Mode 4 is associated with positive precipitation anomalies in the southeastern United States and the west coast of Canada and negative anomalies in the northwestern United States (Fig. 5g).

The CAM5-GOGA qualitatively reproduces the precipitation responses to the four subseasonal modes. In Mode 1, for instance, the model successfully captures the pronounced precipitation decrease confined to the west coast of North America (Fig. 5b). Its response appears slightly stronger than that in the reanalysis data, which can be attributed to the more intense convergence in the southeastern flanks of the anticyclonic circulation in the model. The moderate decrease in precipitation over the western part of North America is also well simulated in the model, coinciding with the southward winds driven by circulation changes. The anticyclonic circulation of Mode 2 explains the dipole structure of precipitation anomaly in North America in the model (Fig. 5d). The precipitation patterns are overall similar to the reanalysis but relatively weaker due to the attenuated circulation in the model. The model also captures the precipitation response in Mode 3, characterized by the pronounced decrease in the northern United States and southern Canada but increase in the northwest coast of North America (Fig. 5f). Lastly, qualitatively similar precipitation changes by Mode 4 are confirmed in the model (Fig. 5h). However,

the precipitation response over the southern United States is slightly shifted poleward compared to the reanalysis.

In summary, the surface responses are overall in line with the corresponding large-scale circulation anomalies. Cold and warm anomalies appear over cyclonic and anticyclonic circulation anomalies. Similarly, drying is found in the eastern (western) part of anticyclones (cyclones), with wetting occurring on the opposite sides. Although slight biases are observed, the model exhibits similar surface responses to subseasonal modes over North America as in the reanalysis.

3.2 Dynamics of the subseasonal variability and its representation in CAM5

In this section, we assess the dynamics of subseasonal modes in the context of the scale-resolving QG geopotential tendency diagnosis (Methods). Since 6-hourly variables required for the diagnosis are not available from CAM5-GOGA, we performed additional experiments with different SST datasets. Although different datasets are adopted, qualitatively similar subseasonal modes are identified as in CAM5-GOGA (Fig. S1). It is also noted that only the dynamics in the reanalysis are presented in the main text, as the model exhibits subseasonal dynamics similar to the reanalysis. Results from the CAM5 are included as supplementary figures (Figs. S2-5), and their differences between the model and reanalysis are highlighted later.

Figure 6 displays the temporal evolution of Mode 1 through regressions of the Z500 anomaly (contour) and Z500 tendencies (shadings) attributed to the QG geopotential tendency equation (Eq. 1) onto the corresponding subseasonal mode index. At lag -3 days of onset, a positive Z500 anomaly emerges over the North Pacific high latitudes and Alaska (Fig. 6a). It develops quasi-stationarily (shading in Fig. 6a), forming a meridional dipole structure. After the onset, the positive anomaly moves slightly westward (Branstator 1987; Kushnir 1987; Fig. 6b), followed by a weakening over the North Pacific at lag 3 days (Fig. 6c). The total tendency is found to be largely explained by the tendency from vorticity fluxes (Figs. 6g–i). In contrast, the tendency driven by heat fluxes exhibits a relatively weaker magnitude compared to vorticity fluxes (e.g., Lau and Holopainen 1984; Lau and Nath 1991). It is characterized by a negative tendency upstream of the anticyclonic circulation but a positive tendency downstream of the circulations (Figs. 6d,e) (e.g., Kug and Jin 2009; Teubler and Riemer 2021). A multi-level assessment further suggests that the contributions of vorticity fluxes in the lower troposphere overall align with those in the upper troposphere, explaining the life cycle of Mode 1 (Fig. S6g–i). In contrast, the contribution of heat fluxes is stronger in the lower troposphere compared to the mid-troposphere and exhibits an opposite sign in the upper troposphere (Fig. S6d–f), consistent with Lau and Holopainen (1984). Although heat fluxes become more important in the lower troposphere, the overall dynamics are explained by vorticity fluxes in Mode 1. Similar results are also found in Modes 2–4 (Figs. S7-9).

Given the important role of the vorticity fluxes, they are further decomposed into high-frequency (HFV) and low-frequency (LFV) vorticity components (Eq. 3). The positive Z500 tendency attributed to HFV appears in North Pacific at lag -3 days of onset (Fig. 6j). It continues to fuel the development of the anticyclonic circulation at onset (Fig. 6k). The HFV height tendency becomes weakened at lag 3 days of the onset, showing a positive tendency along the westward-tilted anticyclonic circulation (Fig. 6l). The Z500 tendency obtained by LFV also contributes to the quasi-stationary amplification of Mode 1 at lag -3 days (Fig. 6m). At the onset, the attributed LFV tendency leads to the westward propagation of Mode 1 (Fig. 6n), followed by a pronounced negative tendency at the North Pacific high latitudes (Fig. 6o). This result indicates that both LFV and HFV play an important role in the formation of Mode 1 with LFV being slightly stronger. On the other hand, the stronger contribution of HFV compared to that of LFV after the onset suggests that HFV is more pivotal in maintaining the system (see Figs. 6k,n). Physically, the amplification of low-frequency (e.g., planetary-scale) eddies prior to the onset of Mode 1 creates favorable conditions for self-interactions among high-frequency eddies (e.g., synoptic systems) and interactions between low- and high-frequency eddies, by anchoring migratory synoptic systems when they pass through the region preconditioned by the low-frequency eddies (Colucci 2001; Nakamura and Huang 2018). The resulting upscale-energy cascade from high-frequency eddy feedback develops and maintains Mode 1 anomalies throughout the life cycles of the system, suggesting distinct roles played by high-frequency and low-frequency dynamics.

The mechanism of Mode 2 is investigated in Fig. 7. An incipient positive Z500 anomaly appears over northeastern North America at lag -3 days, which amplifies quasi-stationary partly due to a Rossby wave train from the North Pacific (Fig. 7a). The positive tendency over the North Pacific suggests a westward expansion of the anomaly at the onset (Fig. 7b; Woollings et al. 2023). After the onset, the westward-expanded anomaly decays due to the negative tendency embedding the wave train (Fig. 7c). Overall tendency is largely accounted for by vorticity fluxes. Although weak, the negative tendency attributed to heat fluxes is located upstream of the anomaly (Figs. 7d-f). It partly counteracts the positive tendency from vorticity fluxes, thereby assisting in maintaining its structure (e.g., Kug and Jin 2009). The attributed tendency by vorticity fluxes is further separated

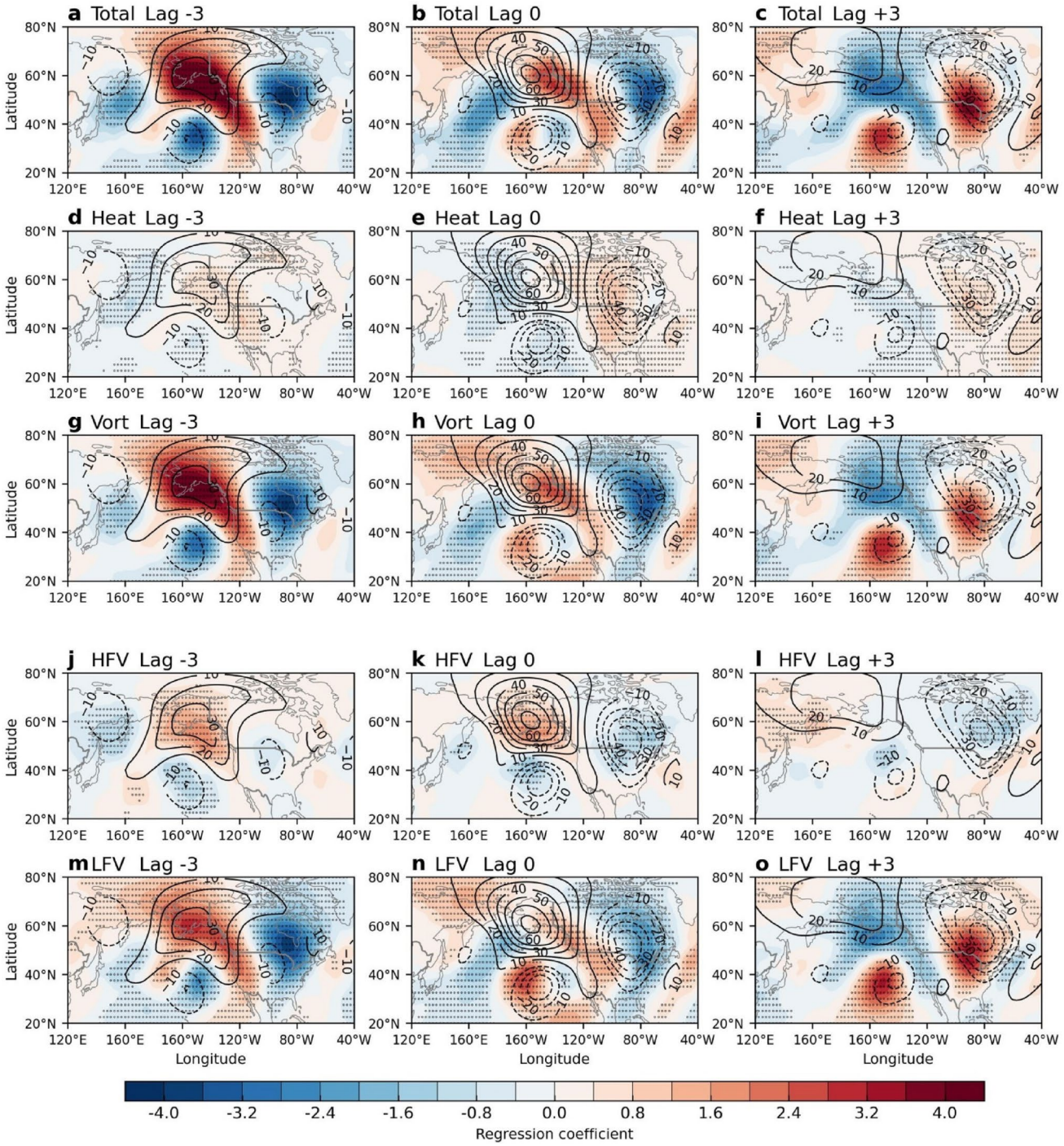


Fig. 6 Regression coefficients of Z500 anomaly (contour) and the attributed Z500 tendency (shading) with respect to subseasonal mode index for Mode 1 from lag -3 to 3 of the onset in ERA-I. The Z500 tendencies attributed to **a**–**c** total, **d**–**f** heat, and **g**–**i** vorticity forcings are displayed in the upper panel. The lower panel shows the attributed tendencies obtained from the vorticity forcings associated with **j**–**l** high-frequency (HFV) and **m**–**o** low-frequency (LFV) components (Methods). The zero line is omitted. The attributed tendencies are 1–2–1 smoothed before calculations. The regions with statistical significance at the 95% confidence level are dotted

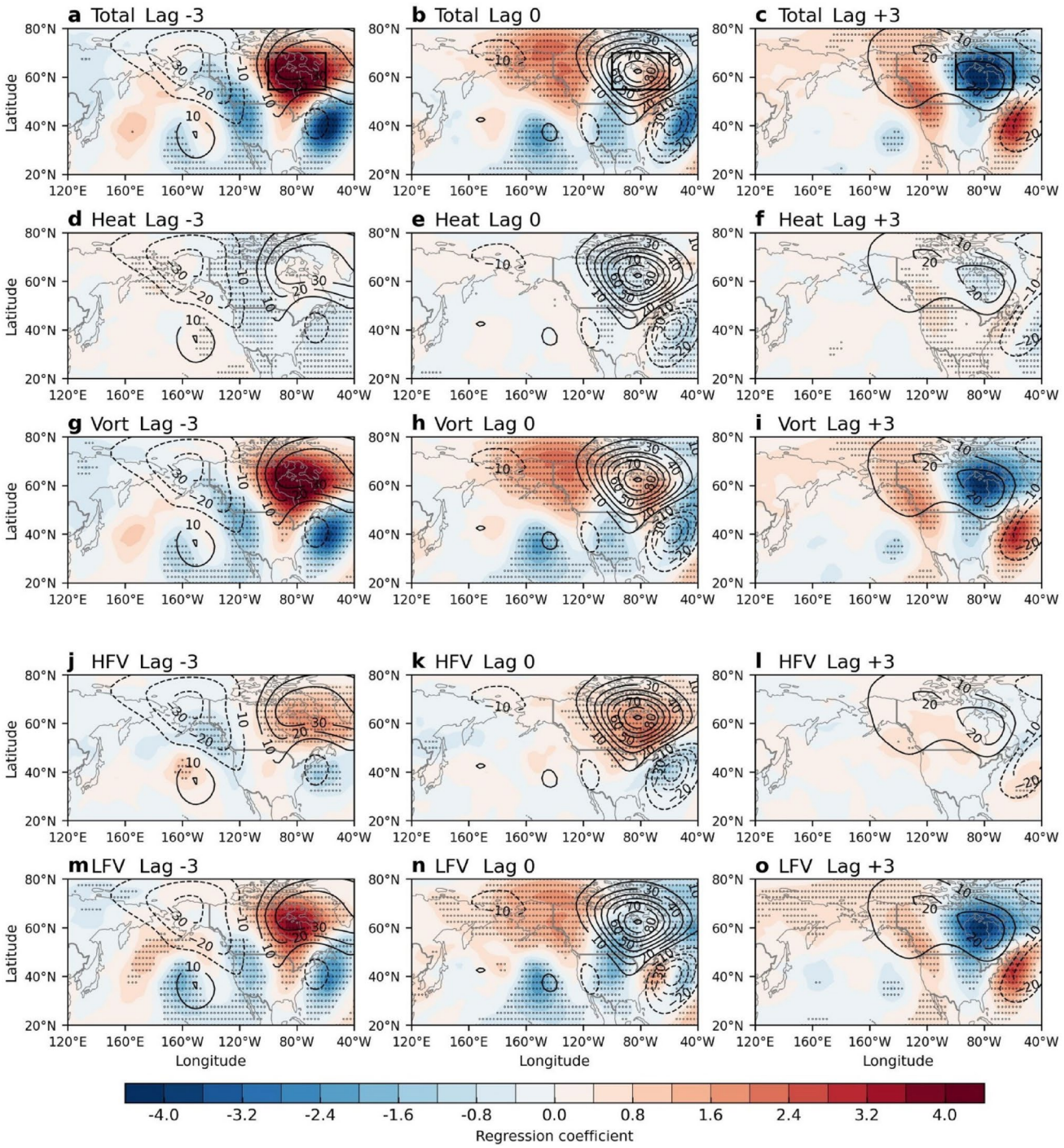


Fig. 7 Same as Fig. 6 but for Mode 2

into HFV and LFV. The height tendencies from both HFV and LFV assist a quasi-stationary development of Mode 2 at lag -3 days (Figs. 7j,m). At the onset, HFV contributes to the system being persistent (Fig. 7k), while LFV displaces the anomalous anticyclone westward (Fig. 7n). After the onset, the decay of Mode 2 is dominated by the negative tendency by LFV (Fig. 7o).

Figure 8 displays the time evolution of Mode 3. Unlike the quasi-stationary development in Modes 1 and 2, the height tendency in Mode 3, which originates from lower latitudes of the North Pacific, is out of phase with Z500 anomalies throughout the life cycle (Figs. 8a–c). The relative position of the two is consistent with the eastward propagation of Z500 anomalies. The height tendency attributed

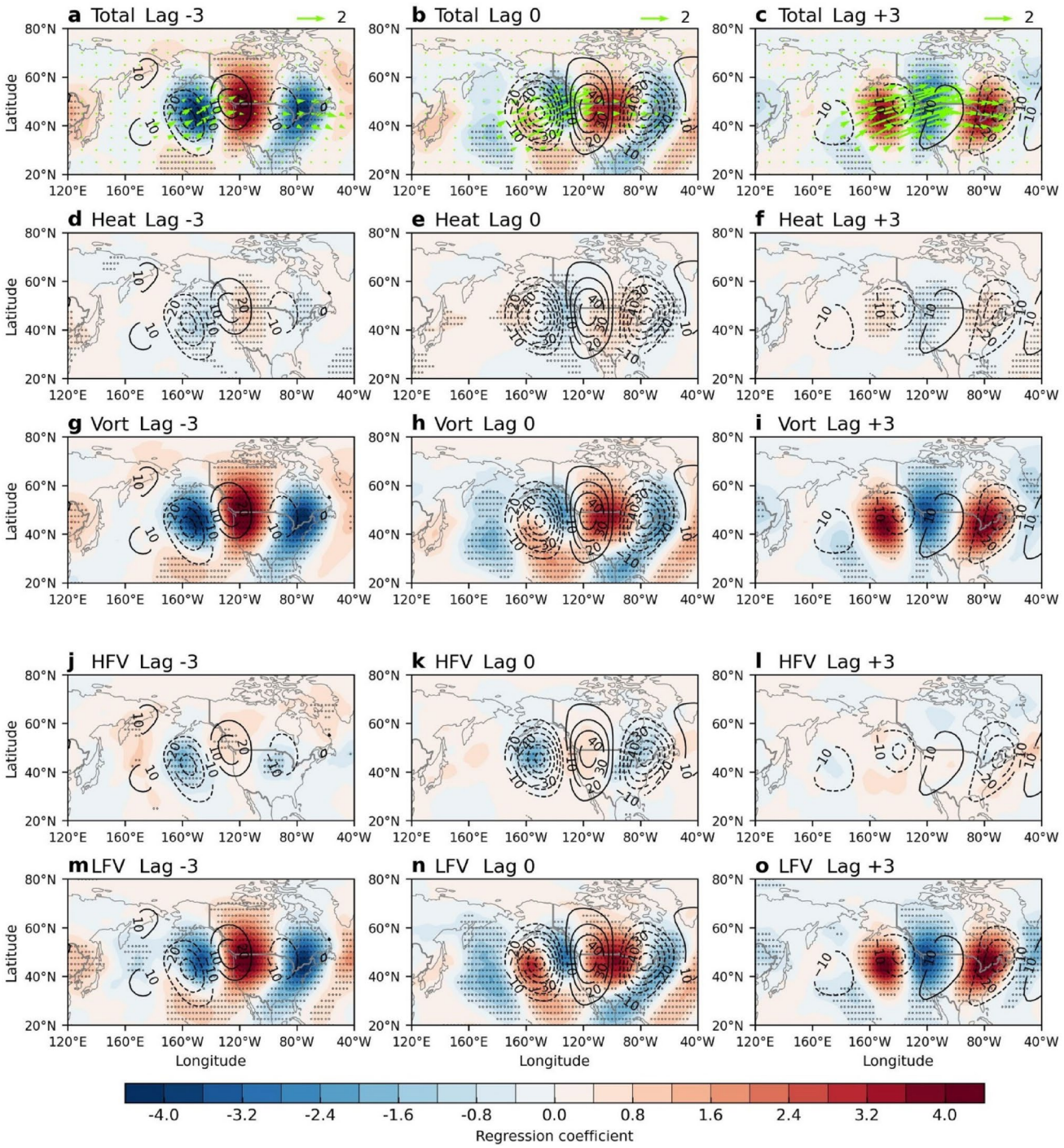


Fig. 8 Same as Fig. 6 but for Mode 3. Vectors indicate wave activity fluxes defined by Takaya and Nakamura (2001)

to both heat and vorticity fluxes exhibits similar patterns, albeit with different amplitudes (Figs. 8d–i). It differs from Modes 1 and 2, where the two fluxes do not always align. The decompositions of vorticity fluxes further show that the eastward propagating nature of the system is driven by LFV (Figs. 8m-o), whereas HFV maintains the wave structure (Figs. 8j–l).

The same analysis is also performed for Mode 4 (Fig. 9). As in Mode 3, the Z500 anomaly and Z500 tendency are out of phase, which explains the eastward propagation of the system (Figs. 9a–c). However, the wave train is more zonally oriented in mid-latitude compared to that of Mode 3. The vorticity fluxes dominate the total tendency in Mode 4 (Figs. 9g–i). The heat fluxes also contribute but are weaker

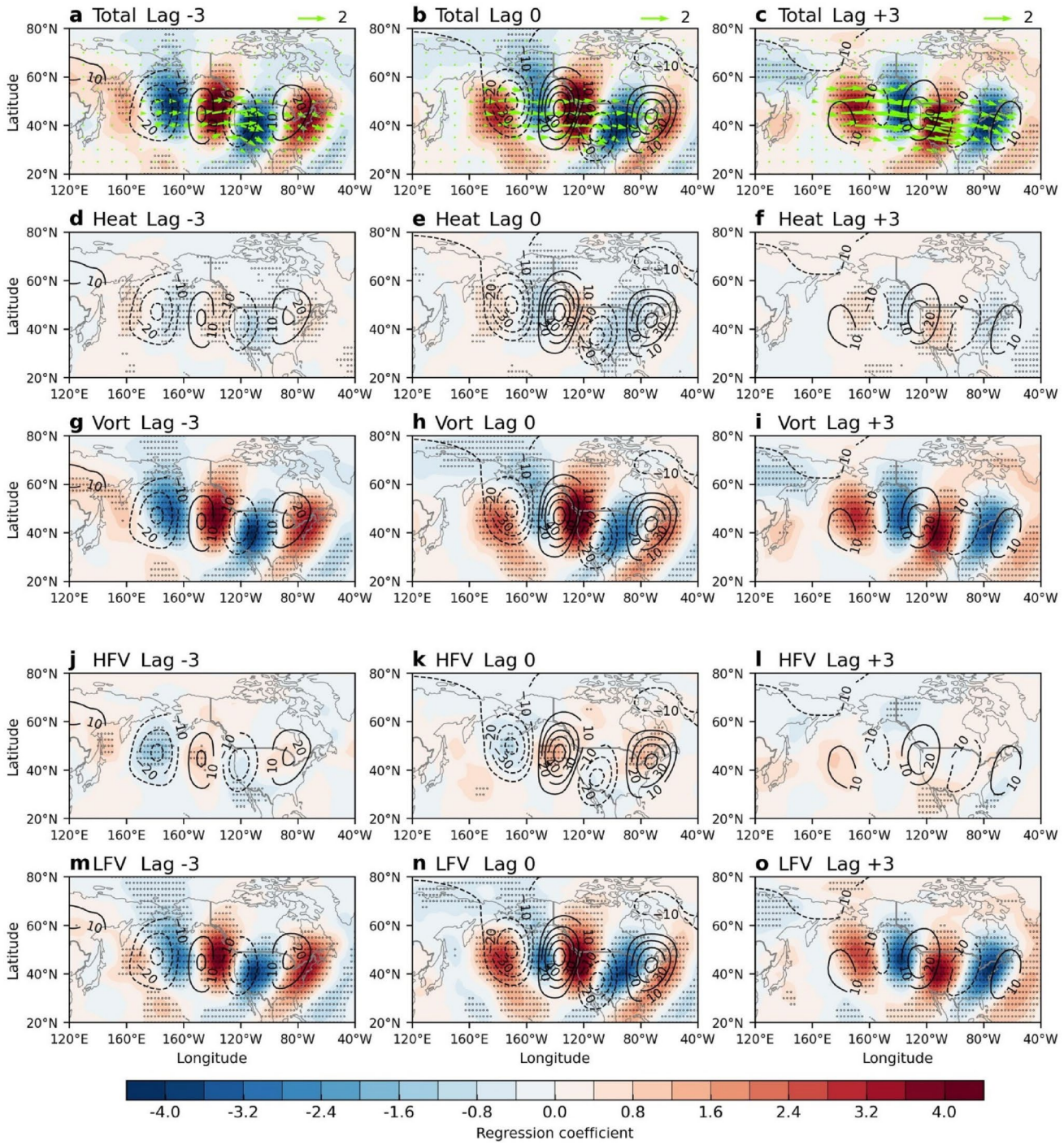


Fig. 9 Same as Fig. 6 but for Mode 4. Vectors indicate wave activity fluxes defined by Takaya and Nakamura (2001)

compared to the vorticity fluxes (Figs. 9d–f). The contribution of the vorticity fluxes to total tendency is dominated by the tendency associated with LFV, explaining the propagating nature of Mode 4 (Figs. 9m–o). The tendency driven by HFV is in line with the Z500 anomaly, maintaining the wave train (Figs. 9j–l).

The above results suggest that LFV plays a major role in the dynamics of the subseasonal modes. However, its role in the formation of these modes noticeably differs between localized high-latitude variabilities and wave-like mid-latitude variabilities. For instance, in the high-latitude variabilities, represented by Modes 1 and 2, the LFV drives a quasi-stationary amplification of the systems. In Modes

3 and 4, both of which exhibit a mid-latitude wave-like structure, LFV amplifies and drives an eastward propagation of the system. In this regard, to elucidate the role of LFV in formation of the subseasonal modes, LFV is further decomposed into the linear- (LFV_l) and non-linear (LFV_{nl}) components.

$$LFV = LFV_l + LFV_{nl} \tag{4}$$

$$LFV_l = -(\nabla \cdot \boldsymbol{V}_c \zeta_{La})' - (\nabla \cdot \boldsymbol{V}_{La} \zeta_c)' - (\nabla \cdot \boldsymbol{V}_{La} f)'$$
$$LFV_{nl} = -(\nabla \cdot \boldsymbol{V}_{La} \zeta_{La})'$$

where subscripts *c* indicates the climatological component. The difference between low-frequency and climatological components is denoted by the subscript *La*.

Figure 10 shows the Z500 tendency driven by the linear and non-linear components of LFV in the four subseasonal modes 3 days prior to onset. In Mode 1, the height tendency from the linear component of LFV resembles the wave train-like pattern which originates from the lower-latitude North Pacific (Fig. 10b). In contrast, the pronounced positive tendency over the high-latitude North Pacific is displayed by the non-linear component of LFV, which explains the quasi-stationary developments of Mode 1 (see Figs. 10a,c). Likewise, dynamics of Mode 2 are also examined (Figs. 10d-f). The linear component of LFV marginally assists the

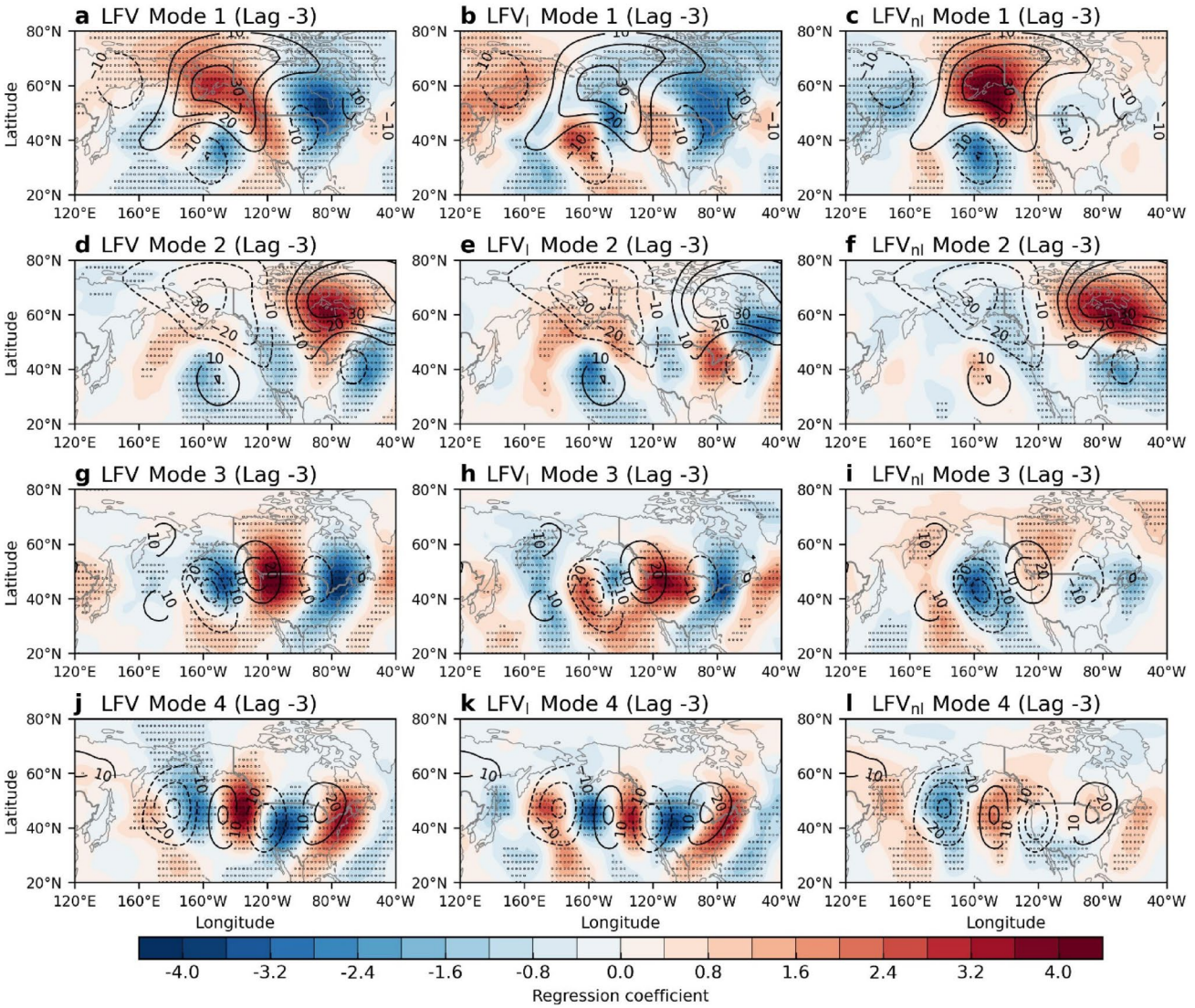


Fig. 10 Regression coefficients of Z500 anomaly (contour) and attributed Z500 tendency (shading) associated with low-frequency vorticity forcing and its decompositions in ERA-I. Results are shown with respect to subseasonal indices for the four modes at lag -3 days of onset. The low-frequency vorticity forcing (LFV; left column) is decomposed as its linear (LFV_l; middle column) and non-linear (LFV_{nl}; right column) components. The zero contour is omitted. The regions with statistical significance at the 95% confidence level are dotted

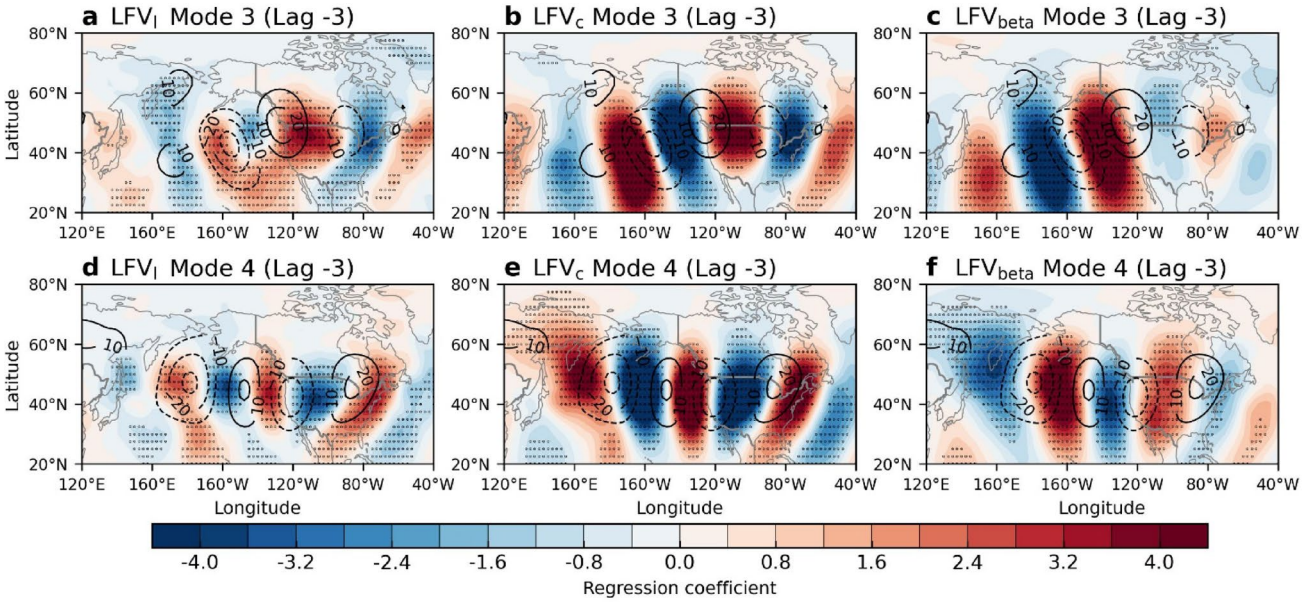


Fig. 11 Same as Fig. 10 but for the Z500 tendency attributed to the linear components of low-frequency vorticity forcing (LFV_l; left column) and its decompositions separated into climatological components (LFV_c; middle column) and beta effect (LFV_{beta}; right column) for Modes 3 and 4

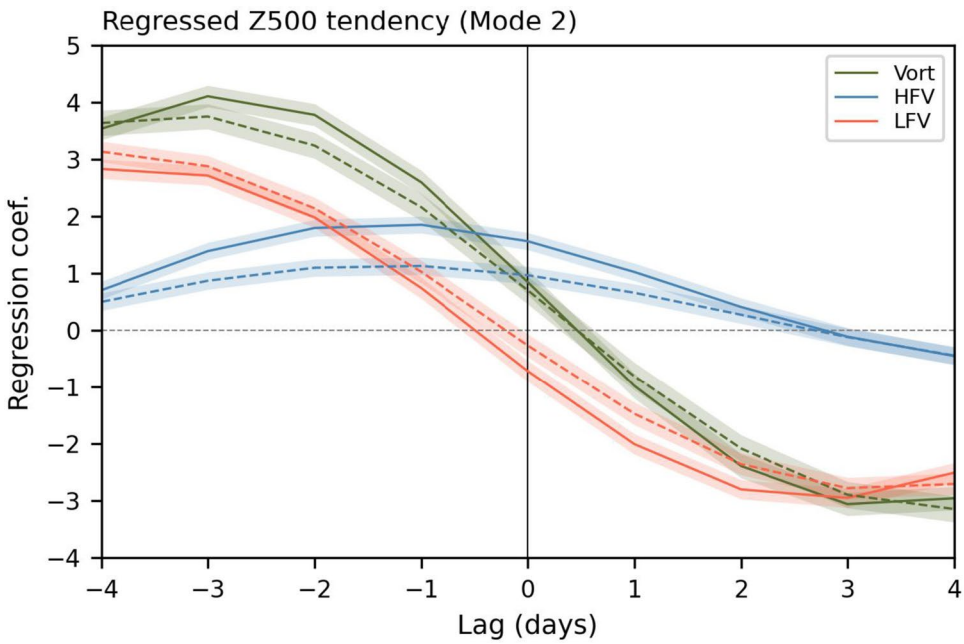


Fig. 12 Time evolution of regression coefficients of the attributed Z500 tendency (shading) with respect to subseasonal mode index for Mode 2 in ERA-I (solid) and CAM5 (dashed), focusing on the region where the maximum amplitude appears (100–60° W; 55–70° N; see boxed region in Fig. 7). Green, blue, and orange lines represent the regression coefficient associated with total vorticity, HFV and LFV forcings, respectively. Shades indicate the 90-percentile confidence interval

development of Mode 2 (Fig. 10e), whereas the non-linear component is in phase, thus dominating the development of Mode 2 (Fig. 10f). In Modes 3 and 4, however, most of LFV originates from its linear component, which drives the eastward propagation (Figs. 10h,k). The non-linear component is much weaker and acts to maintain (or amplify) the wave structure (Figs. 10i,l).

To further understand the contribution of linear component of LFV (LFV_l) to Modes 3 and 4's formation, the height tendencies from the linear component, are decomposed into a component associated with climatological winds and vorticity ($LFV_c = -(\nabla \cdot (\boldsymbol{V}_c \zeta_{La} + \boldsymbol{V}_{La} \zeta_c))'$) and a component due to the beta effect ($LFV_{beta} = -(\nabla \cdot \boldsymbol{V}_{La} f)'$). It is found that the two components have similar structures but opposite signs. By offsetting each other, their sum (i.e., LFV_l) has a much smaller amplitude compared to either individual component. In both Modes 3 and 4, LFV_c (Fig. 11b,e) aligns with their eastward propagating nature (Fig. 11a,d; see also Fig. 10g,j), whereas the beta effects tend to propagate the waves westward (Fig. 11c,f). This result suggests

that Modes 3 and 4 follow the conventional Rossby wave dispersion relation in mid-latitude.

3.3 Tracing the source of Mode 2 bias in CAM5

The dynamics of subseasonal modes, examined by QG geopotential tendency diagnosis, suggest that the four subseasonal modes over North America have their distinct dynamical footprints. While CAM5 qualitatively reproduces the dynamics of subseasonal modes as in the reanalysis (see Figs. S2-5), a substantially weaker Mode 2 is found (see Figs. 2c,d). To identify the potential source of biases, temporal evolutions of the Z500 tendency of Mode 2 in the reanalysis and CAM5 are compared in Fig. 12, by focusing on the regions where maximum amplitude is observed (see boxed region in Fig. 7). The attributed Z500 tendency by vorticity fluxes reveals that the positive tendency peaks at lag -3, followed by a gradual decrease (green lines). The tendency from vorticity fluxes becomes negative after the onset, resulting in a decay of the system. This tendency overall aligns with the tendency from LFV (orange lines), indicating its important role. In contrast, the tendency attributed to HFV exhibits a parabolic-shaped time evolution throughout the life cycle, which acts to maintain the system (blue lines). The model exhibits similar dynamics in Mode 2 (dashed lines). However, the reduced Z500 tendency from vorticity fluxes is found before the onset (green dashed lines), accounting for the weaker amplitude of Mode 2 (see Figs. 2c,d). The tendency attributed to LFV is comparable or even larger in the model compared to the reanalysis (orange dashed and solid lines), whereas a notably reduced contribution from HFV is found in the model (blue dashed and solid lines). Further decomposition of HFV reveals that both the scale interactions between high- and low-frequency eddies and the self-interactions among high-frequency eddies contribute to the attenuated tendency from HFV in the model (Fig. S10). It contrasts with Mode 1 in the model, which shows slightly weaker but largely similar contributions from both HFV and LFV, leading to circulation amplitudes comparable to those in the reanalysis (Figs. 6 and S2).

3.4 Representation of the modes of subseasonal variability in CMIP6 models

To understand the model representations of the four subseasonal modes in fully-coupled models, the statistics of each mode are assessed in historical simulations of 20 CMIP6 models (Table 1). Figure 13 displays the composite of 8–30-day bandpass-filtered Z500 associated with four subseasonal modes in the CMIP6 multi-model mean. The CMIP6 models well capture the circulation patterns and intensities across the four subseasonal modes as in the reanalysis. For Mode 1, both the magnitude and spatial features are remarkably similar to those in reanalysis (Fig. 13a). Like CAM5, the multi-model mean of CMIP6 also exhibits a weaker Z500 anomaly over northeastern North America in Mode 2 (Fig. 13b). However, the bias is less pronounced compared to CAM5. In Modes 3 and 4, the minor amplitude biases are found around the centers of action (Figs. 13c,d). However,

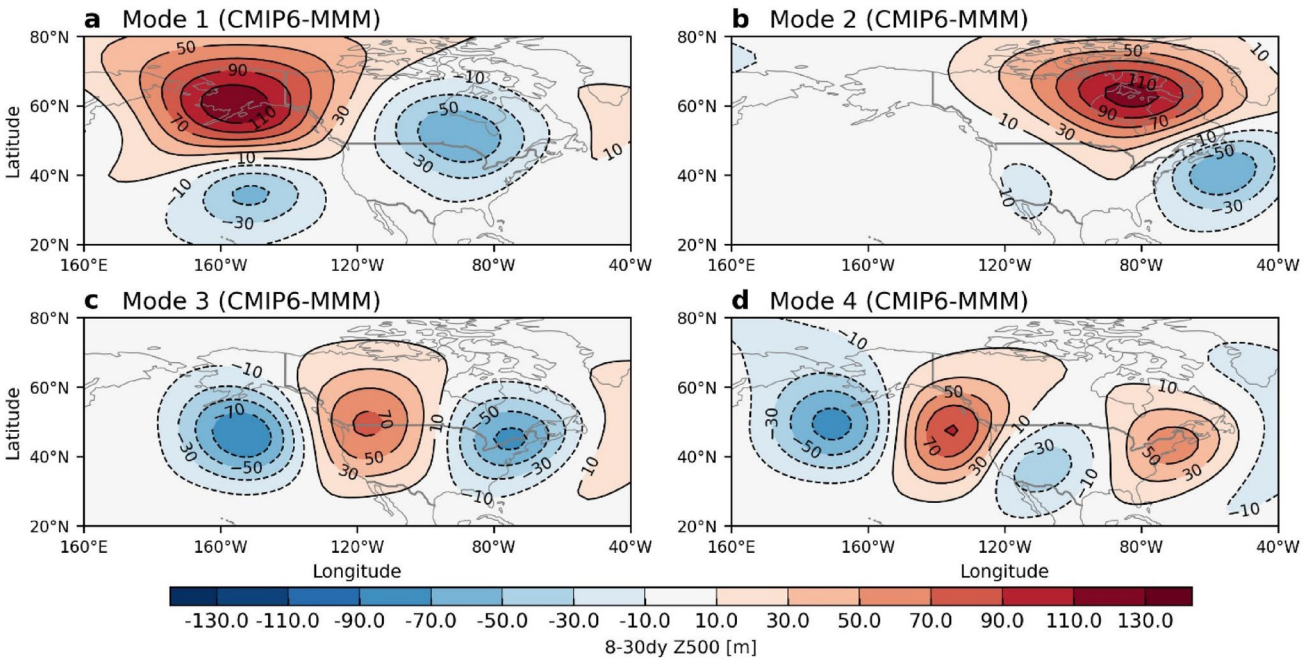


Fig. 13 Same as Fig. 2 but for multi-model mean of CMIP6 (CMIP6-MMM)

there is still a good overall agreement between CMIP6 models and reanalysis.

Figure 14 shows the PDF distributions of the four subseasonal mode indices in CMIP6 models. The PDFs in CMIP6 models exhibit overall similar distributions to those in reanalysis across the subseasonal modes (red lines in Fig. 14). As in CAM5, the distribution of Mode 2 exhibits higher kurtosis in CMIP6 than that in reanalysis (Fig. 14b), in agreement with the attenuated variability of Mode 2 in the models. However, the CMIP6 multi-model mean represents the Mode 2 distribution more accurately than CAM5, which is partly due to the fact that individual model biases cancel each other out. The inter-model spread of CMIP6 is also found to be substantially larger than the single model ensemble spread of CAM5, suggesting that such inter-model spread is likely due to model differences rather than internal variability.

The surface impacts of subseasonal modes in CMIP6 models are also evaluated. Figure 15 displays the changes in surface air temperature and precipitation anomalies associated with each mode in CMIP6 models. As in CAM5, the models capture the overall surface responses to subseasonal modes. However, in contrast to the relatively weak circulation biases in CMIP6, the biases in surface responses are substantial. For instance, in Mode 1, a weakened cold anomaly (~40%) appears downstream of the anticyclonic circulation (Fig. 15a), and a reduced dry response (~30%) is simulated over the west coast of North America compared to the reanalysis (Fig. 15b). Mode 2 induces a weaker warm anomaly (up to ~ 50%) over northeastern North America with a cold anomaly over the western United States in CMIP6 (Fig. 15c). The zonally dipolar precipitation patterns across the northern United States are present for Mode 2 in CMIP6 with 40% diminished amplitudes compared to the reanalysis (Fig. 15d). In Mode 3, the models also reproduce the dipolar temperature anomalies (20–40% weaker) and decreased precipitation (~ 35% less) across the northern United States and southern Canada (Figs. 15e,f) than reanalysis. Similarly, the models exhibit attenuated tripolar temperature anomalies (up to ~ 50% over the western United States) by Mode 4 across North America (Fig. 15g). The increased precipitation response in the downstream region of the cyclonic circulation is captured by models with only 50% of the observed amplitude (Fig. 15h). Notably, low inter-model agreement is also found in the region, indicating a large model uncertainty.

4 Summary and discussion

The present study examines the statistical and dynamical characteristics of springtime subseasonal variabilities over North America, focusing on the four leading modes identified through EOF analysis. Mode 1 is characterized by a strong anticyclonic anomaly over the high-latitude North Pacific with a meridional dipole structure, inducing cold anomalies over North America and reduced precipitation along the west coast. Mode 2 features a localized anticyclonic

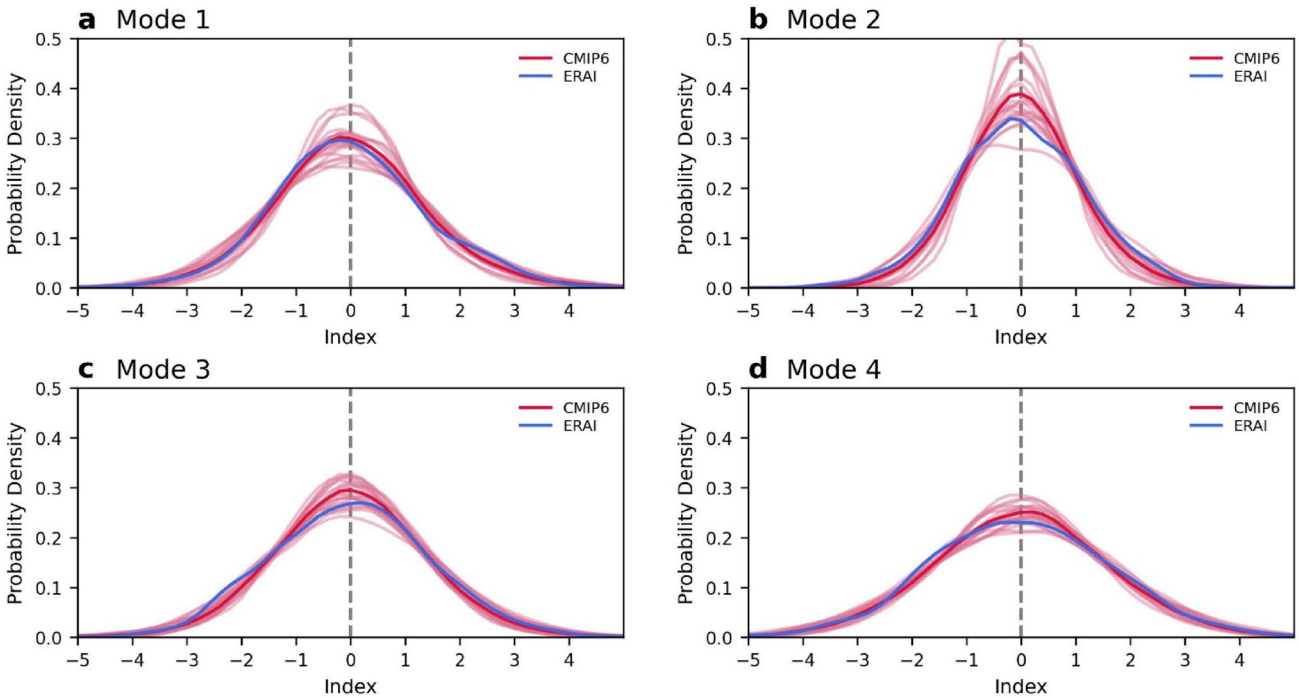


Fig. 14 Same as Fig. 3 but for CMIP6 multi-model mean. The PDF of individual models are illustrated in light red lines

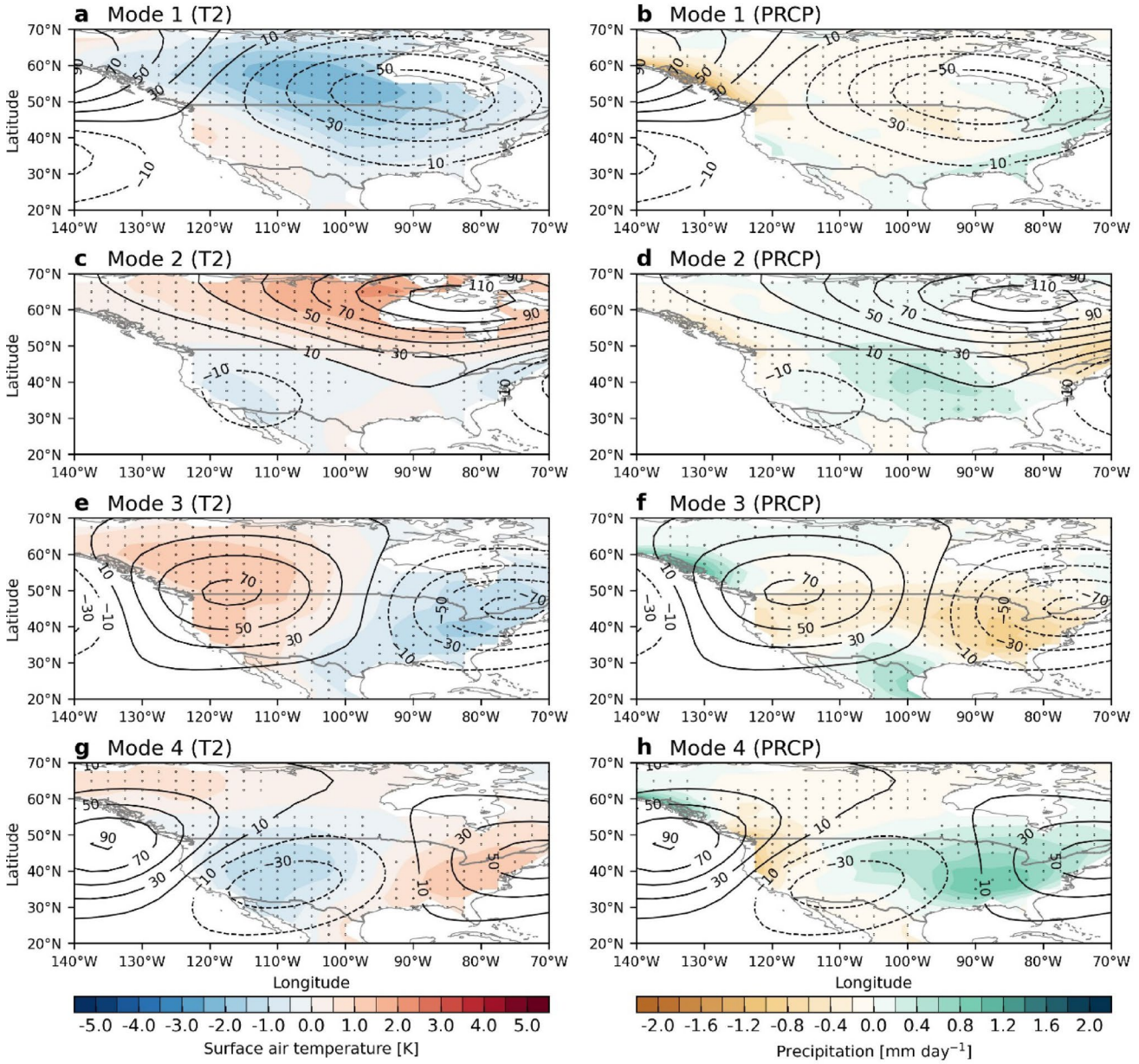


Fig. 15 Same as Figs. 4 and 5 but for CMIP6-MMM. The regions, where more than 2/3 of models show the same sign, are dotted. Here, precipitation in BCC-CSM2-MR is excluded due to data availability

circulation over northeastern North America, accompanied by a warm anomaly over the circulation center with a dipole precipitation anomaly over its upstream and downstream. Modes 3 and 4 both exhibit wave-like patterns spanning from the eastern North Pacific to North America. Mode 3 likely originates from the lower-latitude North Pacific, leading to zonally dipolar temperature responses across North America with a pronounced precipitation decrease over the northern United States and southern Canada. Mode 4 is zonally confined in mid-latitudes. It produces warm-cold-warm temperature anomalies across North America and increases precipitation over the Gulf States.

The dynamics of these subseasonal modes are investigated through a scale-resolving quasi-geostrophic geostrophic tendency diagnosis. Both Mode 1 and Mode 2 develop in a quasi-stationary fashion, showing a meridional dipole structure, followed by a westward movement after the peak intensity of the Z500 anomalies. The height tendency driven by vorticity fluxes dominates the overall evolution of both modes. Although weak, the height tendency attributed to heat fluxes partly compensates for the upstream positive tendency from vorticity fluxes, thereby acting to maintain the structure of the system. By decomposing the vorticity fluxes into high- and low-frequency components,

it is revealed that high-frequency eddy fluxes (e.g., synoptic-scale cyclones and anticyclones) are responsible for the maintenance of the systems, whereas low-frequency eddy fluxes (e.g., pre-amplified planetary waves) explain the growths and decays. In contrast, Modes 3 and 4 exhibit eastward propagation throughout their life cycles, which is driven by both vorticity and heat fluxes. However, the contribution of heat fluxes is much weaker than that of vorticity fluxes. In particular, the eastward propagations are closely related to the low-frequency components of vorticity fluxes. Further decompositions of low-frequency eddy fluxes into linear and non-linear components reveal that the non-linear processes drive the formation of Modes 1 and 2, whereas the linear processes dominate the eastward propagation of Modes 3 and 4 demonstrating the conventional dispersion relation of the Rossby wave.

The statistics and surface responses associated with the four subseasonal modes are evaluated using CAM5-GOGA. While the model exhibits systematic biases in Mode 2, it generally captures the spatial structures of the subseasonal modes and their underlying dynamics. The attenuated variability of Mode 2 in the model is likely due to the weaker high-frequency vorticity fluxes in the model. It is presumably associated with a coarse model resolution or physical parameterizations resulting in biases of smaller-scale processes such as warm conveyor belts and diabatic processes within synoptic systems that are crucial for accurately capturing high-frequency variability (e.g., Jung et al. 2006; Willison et al. 2013; Wang et al. 2023). The model successfully simulates the surface air temperature and precipitation responses to the circulation anomalies of the subseasonal modes, although a poleward-shifted precipitation response appears over the Gulf States in Mode 4.

The statistics of the four subseasonal modes in CMIP6 models are also examined. The CMIP6 multi-model mean well reproduces the distribution and spatial features of the four modes. A notable bias in Mode 2 of CAM5-GOGA is improved in CMIP6 multi-model mean partly due to the compensation of biases from individual models and the potential improvement of air-sea-ice interactions thus smaller-scale synoptic features over high-latitude North America. The latter is further confirmed by the comparison between fully-coupled and SST and sea ice prescribed CAM5 experiments (Fig. S11). Although circulation patterns are well simulated, substantially weaker surface temperature and precipitation responses are found across the subseasonal modes in the CMIP6 multi-model mean. This result indicates that even when subseasonal modes are well simulated by models, biases in surface responses can still be large. This result suggests that representation of precipitating systems (Catto et al. 2013; Chang 2013; Herman and Schumacher 2016; Schumacher and Rasmussen 2020; You et al. 2024) and physics parameterizations (e.g., Pinto et al. 2015; Van Weverberg et al. 2018) need further improvement in the models. Moreover, little consensus appears in the precipitation response to Mode 4 across the CMIP6 models. This is likely due to model diversity in representing interactions between subseasonal and smaller-scale (e.g., synoptic and mesoscale) variabilities, since precipitation over Gulf States is largely related to frontal and/or mesoscale convective systems (e.g., Kunkel et al. 2012; You and Deng 2023; Feng et al. 2021). It underlines the need for evaluating model precipitation in a multi-scale context that synthesizes information from subseasonal to synoptic and mesoscales (e.g., Deng and Jiang 2011; Jiang et al. 2014; Zhang et al. 2021).

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Author contributions J. Hwang conceptualized the idea, conducted the analysis, illustrated the figures, and wrote the manuscript. Yi Deng proposed the overall research idea and contributed to the writing of the manuscript. Zhenyu You and Jie He improved the interpretation of the results and helped to write the manuscript. All of the authors discussed the results and reviewed the manuscript.

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Data availability The ERA-Interim Data and CPC precipitation used in this study were obtained from https://www.ecmwf.int/en/forecasts/datasets/reanalysis-datasets/era-interim and https://psl.noaa.gov/data/gridded/data.cpc.globalprecip.html. The merged Hadley-OI data for CAM5 simulation are available at https://svn-ccsm-inputdata.cgd.ucar.edu/trunk/inputdata/atm/cam/sst/. The CAM-GOGA simulation data are now available at https://www.cesm.ucar.edu/working-groups/climate/simulations/cam5-prescribed-sst. The CMIP6 archives are availabl e from https://aims2.llnl.gov/search/cmip6.

Declarations

Conflict of interest The authors declare no competing interests.

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